Dongha Choi

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RESEARCH INTERESTS

AI-RAN, semantic communications, and real-world wireless prototyping, with an emphasis on translating learning-based communication methods into practical systems.

EDUCATION

2023 - Present **Yonsei University**, Seoul, Republic of Korea
Ph.D. Candidate *(Advisor: Prof. Chan-Byoung Chae)*

2020 - 2022 **Yonsei University**, Seoul, Republic of Korea
B.E. in Integrated Technology *(Early Graduation)*

PUBLICATIONS

Published Manuscripts

1. Hanju Yoo, **Dongha Choi**, Songkuk Kim, Chan-Byoung Chae, and Robert W. Heath, Jr., "Symbol Distributions in Semantic Communications: A Source-Channel Equilibrium Perspective," *IEEE Transactions on Communications*, vol. 74, pp. 13842-13855, 2026.
2. Hanju Yoo, **Dongha Choi**, Yonghwi Kim, Yoontae Kim, Songkuk Kim, Chan-Byoung Chae, and Robert W. Heath, Jr., "Bridging Neural Networks and Wireless Systems with MIMO-OFDM Semantic Communications," *IEEE Wireless Communications*, vol. 32, no. 5, pp. 48-55, 2025.
3. **Dongha Choi** and Chan-Byoung Chae, "A Study on the Transformer-based Equalizer in Semantic Communication Systems," in *Proceedings of the Symposium of the Korean Institute of Communications and Information Sciences*, pp. 962-963, 2024.

Manuscripts Under Review / In Preparation

1. **Dongha Choi**, Jungwoo Kim, Hyeonjin Lee, Yoonjung Lee, Jong-Seok Lee, and Chan-Byoung Chae, "Multimodal Semantic Communications for Physical AI: Shared Context Learning and Prototype Validation," *to be submitted*.
2. **Dongha Choi**, Hanju Yoo, Kyungpil Lee, Dongwook Kim, and Chan-Byoung Chae, "Explainable Semantic Communications: Neural Encoder Interpretation and Prototype Validation," under major revision for a special issue of *IEEE Communications Magazine*.
3. Hanju Yoo, **Dongha Choi**, Songkuk Kim, and Chan-Byoung Chae, "Back to the Bits: Parametric Symbol Compression for Semantic O-RAN Fronthaul," *to be submitted*.
4. **Dongha Choi**, Hanju Yoo, and Chan-Byoung Chae, "Steganography Semantic Communications: Architectures and Prototype Validation," *to be submitted*.

SYSTEM DEMONSTRATIONS

Dec. 8-12, 2025 **IEEE GLOBECOM 2025**, Taipei, Taiwan
Secure Semantic Communications with Steganography

Jun. 15-20, 2025 **IEEE IMS 2025**, San Francisco, USA
Real-Time Semantic Communications with Transparent AR Glasses (with Sensorview)

May 14-15, 2025 **IEEE 6G Summit Dresden 2025**, Dresden, Germany
Real-Time Semantic Communications with Transparent AR Glasses, as part of the Real-Time 6G Wireless Systems exhibition

AWARDS AND HONORS

Feb. 2025 **Honor Prize**, 31st Samsung Humantech Paper Award *(Co-recipient: Hanju Yoo)*

Mar. 2022 **Full Scholarship for Undergraduate**, Kwanjeong Educational Foundation

Aug. 2021 **Highest Honors**, top 1% in the College of Engineering

Feb. 2020 **Best Dreamer Award**, high school valedictorian

EXPERIENCE

Dec. 2024 - Jan. 2025 **Visiting Researcher**, Singapore University of Technology and Design
Hosted by Prof. Jihong Park; collaborative research on semantic communications.

2022 - 2025 **Teaching Assistant**, Yonsei University
Computational Thinking and SW Programming; Engineering Information Processing; Understanding the World with Data
Led weekly Python, C, and SQL sessions and prepared course materials and examinations.

PROFESSIONAL SERVICE

2026 - Present **Reviewer**, IEEE Journal of Selected Topics in Signal Processing

SKILLS

Languages Korean (Native), English (Fluent), Chinese (Intermediate; HSK IV 280)

EECS Communications Theory, Information Theory, Machine Learning, Data Management, Digital System Design, and Computer Systems Programming

Programming Python, TensorFlow, PyTorch, MATLAB, C (Fluent); JavaScript, HTML (Intermediate)